# Midterm Quiz

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# Exam: TV Program Recommendation

## Task

Use the SWEET.TV competition data to produce a ranked Top-5 list of TV programs for each test user.

## Implementation

- Implement at least two methods: Content-Based Filtering (CBF), Association Rules, Matrix Factorization (CF), or Personalized PageRank.
- Connect viewing logs with the program schedule to construct user–program interactions. Explain your matching rule and treatment of missing program IDs.
- Compare methods using the same temporal validation split, candidate pool, and evaluation protocol.
- Include a popularity baseline as a reference; it does not count as one of the two required methods.
- Your final submission may use a single method or a hybrid strategy. Evaluate each selected method individually before combining them.
- Use the provided program metadata, but do not use future user viewing behavior for training.

## Business & Algorithmic Analysis

**Which recommendation strategy would you select for SWEET.TV, and why?**

- Select two users and justify your selection. Use the same users across all implemented methods.
- Show each method's Top-5 with program IDs, titles, and categories. Compare the recommendations with the users' viewing histories and program attributes.
- Explain how and why your selected methods differ, and what those differences imply for the user experience.
- Justify your final strategy using validation performance, computational cost, and specific recommendation examples.
- Discuss limitations and failure cases supported by your results. No live A/B test or demonstrated revenue impact is required.

## Submission Deadline: 24th Oct 19:00

Submit only:

1. Your report in PDF format.
2. Your original, unmodified AI-session export.

The PDF report must include:

- A link to your GitHub repository.
- A screenshot showing the returned Kaggle private score for your designated final submission.
- Your method comparison, validation results, and business and algorithmic analysis. Same like Homework.

The GitHub repository must contain:

- Your implementation code.
- Instructions for reproducing the results. (Optional)
- The final submission file used to obtain the reported Kaggle score.

## Grading

**(Gate0 is automatic fail of exam, please check the homework guideline carefully!)**

**Kaggle leaderboard-score ranking among exam participants — 50 points**

Participants are ranked by the returned score of their designated final submission, not by their position on the frozen Kaggle leaderboard.

Successive 20% groups receive 50 / 40 / 30 / 20 / 10 points.

**Homework-style evaluation — 50 points**

The existing homework evaluation criteria (c1 and c2) apply, covering implementation, the PDF report—including business analysis—and the original, unmodified AI-session export.

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Late Submission ···



# sweet.tv - TV program recommender

Create the algorithm that predicts user's TV program preferences

Click this one!

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

## Overview

**Start**
Nov 18, 2020

**Close**
Nov 22, 2020

### Description

Cinema Challenge Hackathon dedicated to finding solutions for online cinema. The prize pool is $50 000. We invite data scientists, developers, managers, and designers to participate. Participation is possible both individually and as a team, depending on the chosen setting. What awaits the participants? • $10 000 — for first place in every challenge • $5 000 — for second place in every challenge • 500+ participants — useful acquaintances • professional and personal development You are allowed to use any external data source 10 submissions/day Deadline - 21.11.2020, 11.59 PM Your task is to predict the 5 most preferred **tv_show_id**s for each user from submission.csv, from the most preferred (#1) to the least preferred (#5). The preferences were calculated for the users using the subsequent 12 weeks (weeks 31-42) of historical views.

**Competition Host**
ogurbych

**Prizes & Awards**
Kudos
Does not award Points or Medals

**Participation**
69 Entrants
47 Participants
36 Teams
578 Submissions

**Tags**
MAP@{K}

**Table of Contents**
Description

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# sweet.tv - TV program recommender

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## Dataset Description

### File descriptions

- **dataset11-30week.csv** - the training set. **vsetv_id** is the same as **channel_id** in export_arh_*.csv files.
- **export_arh_10-19.csv** - supplemental information about train data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **export_arh_20-30.csv** - supplemental information about train data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **export_arh_31-42.csv** - supplemental information about test data (TV program schedule). Contains tv_show_id - you have to predict 5 of these for the submission.csv user_ids. Note: tv_show_id = 0 stands for irrelevant TV shows, drop them!
- **submission.csv** - a sample submission file in the correct format. Substitute 5 zeros with 5 predicted tv_show_ids for each user!

### Train data fields

- **user_id** - an anonymous id unique to a given user
- **vsetv_id** - the id of a channel (equal to **channel_id** in export_arch_*.csv)
- **start_time** - start time of channel being watched
- **end_time** - end time of channel being watched

Submit to Competition

File Upload Notebook MCP

**sweet.tv - TV program recommender**

You have 99 submissions remaining today. This resets in 9 hours.

Uploaded File

Your submission file

submission.csv (59 KiB)

Your submission should be a CSV file with 1260 rows and a header. You can upload a zip/gz/7z archive.

Submission Description

Write your full name in Russian here!!!!
Джин

Your full name (Russian) & Kaggle name

45 / 500

Submit using the Kaggle CLI

```
>_ kaggle competitions submit -c sweettv-tv-program-recommender -f …
```

Cancel Submit

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Create the algorithm that predicts user's TV program preferences



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## Submissions

You selected 0 of 2 submissions to be evaluated for your final leaderboard score. Since you selected less than 2 submissions, Kaggle auto-selected up to 2 submissions from among your public best-scoring unselected submissions for evaluation. The evaluated submission with the best Private Score is used for your final score.

0/2

Submissions evaluated for final score

All | Successful | Selected | Errors

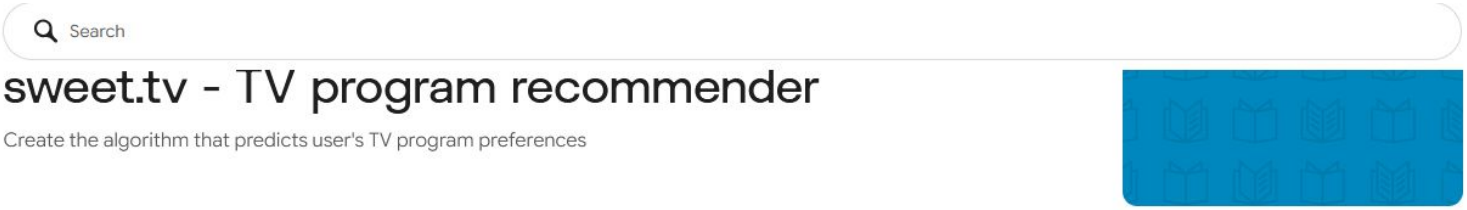
Recent

| Submission and Description | Private Score | Public Score | Selected |
|---|---|---|---|
| submission.csv<br>Complete (after deadline) · now · Джин Сеунгмин | 0.00000 | 0.00000 | ☐ |
| submission.csv<br>Error (after deadline) · 1m ago · Джин Сеунгмин | | | |
| submission.csv<br>Complete (after deadline) · 1h ago · template verification | 0.00000 | 0.00000 | ☐ |

**Error case!!**

Please make sure that your name is displayed correctly.

Number of submission is limited to 5 usually. So be wise.




Capture the your Submission page as a screenshot and attach it into the report.

I will accept the best score among them

If you have any issues, please dm to me.

@joseph_jin